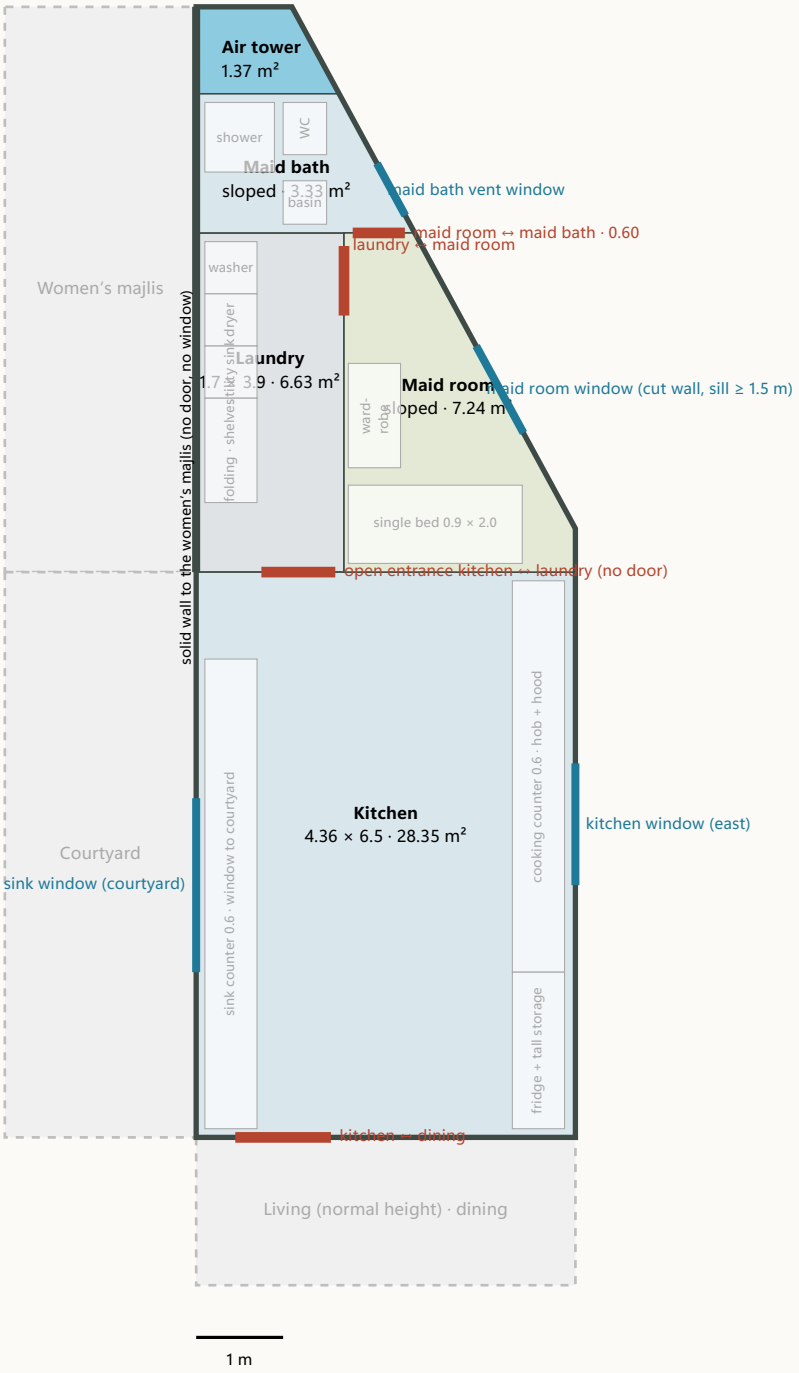
East service block · kitchen, laundry, maid wing, air tower · layout study

Taif / Palm House · Design V3.4 (Z4) · block = CSV x 8.95–13.311, y 0–13.0 (NE corner cut) · 46.91 m² · scale 1 m = 1.15 cm · north up



Rooms

Room	w × d (m)	m²
Maid bath	sloped (to the cut wall)	3.33
Maid room	sloped (to the cut wall)	7.24
Laundry	1.7 × 3.9	6.63
Kitchen	4.36 × 6.5	28.35
Air tower (polygon)	4 corners	1.37
Leftover slivers along the cut		0
Block		**46.91**

✓ **All rooms inside the block, no overlaps**

Route

Family: living (normal height) · dining → kitchen (south door).
Maid: kitchen → laundry → maid room → maid bath. No external door and no other door: all her movement passes through the kitchen.
Cooking: sink on the courtyard wall, hob and hood on the east (outside) wall, so heat and smoke leave the house; fridge and tall storage at the south-east, next to the dining door, both on the new east wall face.

Rules

- Kitchen isolated: doors only to the dining area (south) and the laundry (north). No door to the courtyard, riwaq or women's majlis.
- Kitchen 4.361 × 6.5 m on axes (28.35 m², ≈ 26.1 m² clear, ≈ 4.136 m clear wide), y 6.5–13.0, exactly on the courtyard lines. No island: two parallel counter runs, the sink run on the courtyard wall and the hob, hood and fridge run on the east outside wall; ≈ 2.94 m between the counter fronts. This is the heavy (dirty) kitchen; the hospitality kitchen in the west block is the light one.
- Laundry vertical on the west wall (women's majlis side), 1.70 × 3.9 m (x 0–1.70, y 2.6–6.5), from the kitchen up to the maid bath: open entrance from the kitchen in its south wall (no door); washer, dryer, utility sink and folding shelves in one line on the west wall (solid, no opening, the wet wall 200 mm); ≈ 0.95 m clear beside the 0.6 m machines (1.55 m between the 200 mm wall and the 100 mm partition); maid room door at the north end of its east wall. The maid's only route. No outside wall: needs a mechanical extract fan to the roof.
- Maid room east of the laundry behind a 100 mm dry block partition, from y 2.6 down to y 6.5; the slope ends at y 6.0, then 0.5 m of straight east wall (≈ 7.24 m² on centre lines, ≈ 6.18 m² clear, still under the 6.5 m² target): door from the laundry at the north end of its west wall and door to the maid bath in its north wall, both in the narrow north part; single bed 0.9 × 2.0 along the south wall (the widest part, ≈ 2.4 m clear) and a wardrobe on the west wall; window in the cut wall with a sill ≥ 1.5 m (privacy from the lane).
- Main cross walls on the house lines: y 6.5 (women's majlis / courtyard) and y 13.0 (dining). Exterior walls and the air tower unchanged; all interior walls straight and square. Walls: 100 mm blocks for dry partitions, 150–200 mm for walls carrying pipes (maid bath, laundry sink wall).
- Maid bath 1.6 m deep (y 1.0–2.6), running up to the cut wall (≈ 3.33 m², ≈ 2.57 m² clear); door from the maid room 0.60 m (sliding or opening out): WC, basin, compact shower; vent window in the cut wall; door only from the maid room.
- Solid wall (no door, no window) between the maid wing and the women's majlis.
- Air tower in the tip of the cut corner, 1.0 m deep (≈ 1.37 m² on centre lines, ≈ 0.88 m² clear, at least ≈ 0.93 m wide at its narrow north end): one person can step in to clean; main cleaning from a roof hatch. Takes no room space.
- Plumbing: kitchen, laundry and maid bath drain under the slab straight out to the east, to a manhole in the east service yard; a small east duct (≈ 0.6 × 0.6 m) for the vent stack, any first-floor bathroom above and the kitchen extract; water from the main manifold through the ceiling to a small secondary manifold in the laundry. Coordinate drain depths with the air tunnel (also under the east side).
- Open decision (not drawn): a small service door from the laundry to the east service yard, locked with the owner's key only, for rubbish and deliveries. Gas (if used): cylinder cabinet outside the east wall in the service yard.